

Hack The Web – Tedious Work

In math class, our teacher tortured us with assignments! His specialty was long chain calculations like this one:

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = ?$$

I always dutifully calculated them in order. Back then I couldn't understand how some of my classmates managed to do it in their heads.

Even today, I prefer to use a calculator in such cases. But no matter which method you use: The result of the calculation is your answer.

In this challenge, we're told to do some math,
 $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$

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Adding Consecutive Numbers

In the case that we have to add consecutive numbers, there is a trick that can make manual calculation much quicker



1
2
3
4
5
6
7
8
9
10

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Adding Consecutive Numbers

$$(1 + 10) * (10 / 2) = 55$$

If we add the first and last numbers in the set, then multiply that number by the number of items in the set divided by two, then we will get the correct answer

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Adding Consecutive Numbers

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9$$

$$(1 + 9) * (8 / 2) + 5 = 45$$

In the case there are an odd number of items in the set of consecutive numbers, we remove the number in the middle, and add it at the end